SHEET 1 – **OPTIMIZATION FOR DATA SCIENCE** SS 2026
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In data science one is often interested in optimization problems of the form

$$\min_{x \in \mathbb{R}^n} f(x)$$

with

$$f(x) := \frac{1}{m} \sum_{i=1}^{m} f_i(x).$$

Examples include:

- linear least squares regression:
  Given a data set $\mathcal{D} = \{(a_i, b_i) \in \mathbb{R}^k \times \mathbb{R} \ : \ i \in \{1, \dots, m\}\}$ (a_i is typically called *feature*, b_i is called *observation* or *label*), the functions $f_i : \mathbb{R}^n \to \mathbb{R}$ are, e.g., for least squares minimization with intercept 0 given by

  $$f_i(x) := \frac{1}{2}(a_i^\top x - b_i)^2$$

  with $k = n$.

- variants add regularization terms, e.g., ridge regression adds an ℓ^2-regularization

  $$f_i(x) := \frac{1}{2}(a_i^\top x - b_i)^2 + \frac{\lambda}{2}\|x\|_2^2,$$

  or the so-called LASSO formulation adds an ℓ^1-regularization (that promotes sparsity of the solution)

  $$f_i(x) := \frac{1}{2}(a_i^\top x - b_i)^2 + \lambda\|x\|_1,$$

  or replace the mapping $x \mapsto a_i^\top x$ by $x \mapsto h(a_i, x)$, where $h : \mathbb{R}^k \times \mathbb{R}^n \to \mathbb{R}$.

Questions:

- For which of the above applications is f convex, strongly convex, smooth?
- Verify that for linear least squares regression the objective function can be written as $f(x) = \frac{1}{2m}\|Ax - b\|_2^2$, where the rows of $A \in \mathbb{R}^{m \times n}$ are given by $a_i^\top$ and $b := (b_1, \dots, b_m)^\top$.

Exercise 1.1

Install Jupyter on your laptop and test the installation by running the jupyter notebook accompanying the exercise sheet.
Option 1:

1. Open a terminal
2. Create a folder `mkdir your-folder-name`
3. Navigate to your folder: `cd your-folder-name`
4. Create a Python virtual environment: e.g., by `python -m venv your-venv-name`
5. Activate the environment: `source your-venv-name/bin/activate`
6. Install packages:
   ```
   pip install matplotlib
   pip install ipympl
   pip install ipykernel
   pip install jupyter notebook
   ```
7. Download `ExerciseSheet1.ipynb` and move it to `your-folder-name`
8. Run `jupyter notebook`

Option 2:

1. Install UV by following the steps on `https://docs.astral.sh/uv/`.
2. Open a terminal
3. Create a folder `mkdir your-folder-name`
4. Navigate to your folder: `cd your-folder-name`
5. Initialize UV environment: `uv init`
6. Install packages:
   ```
   uv add matplotlib
   uv add ipympl
   uv add ipykernel
   uv add jupyter notebook
   ```
7. Download `ExerciseSheet1.ipynb` and move it to `your-folder-name`
8. Run `uv run jupyter notebook`

Exercise 1.2

Let $n, m > 0$, $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $f : \mathbb{R}^n \to \mathbb{R}$ given by

$$f(x) := \frac{1}{2}\|Ax - b\|_2^2.$$

Compute the gradient ∇f and the Hessian $\nabla^2 f$ of f. Is the Hessian $\nabla^2 f$ positive semi-definite? Is f L-smooth?

Exercise 1.3

Let $n \in \mathbb{N}$ and $f : \mathbb{R}^n \to \mathbb{R}$ be L-smooth. Show that for arbitrary $x, d \in \mathbb{R}^n$ there holds

$$f(x+d) - f(x) \leq \nabla f(x)^\top d + \frac{L}{2}\|d\|_2^2.$$

Exercise 1.4

Let $X \subset \mathbb{R}^n$ open and convex and $f : X \to \mathbb{R}$ be continuously differentiable. Then f convex if and only if for all $x, y \in X$ we have $\nabla f(y)^\top (x - y) \leq f(x) - f(y)$.